Week 6 Quiz Chapter 1

Date:

M T W T F S S

Q. $f(x) = \begin{cases} \frac{x^2}{a} - a & \text{if } 0 < x < a \\ 0 & \text{if } x = a \\ a - \frac{a^2}{x} & \text{if } x > a \end{cases}$

at $x = a$

$\lim_{x \to a^-} \frac{x^2}{a} - a = 0$ Continuous

$\lim_{x \to a^+} a - \frac{a^2}{x} = 0$

ME-22101 EE-24159 CE-22191
ME-25152 CE-24028 ME-25152
CE-25211 CE-24030 CE-25211

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Chapter 2:- Differential Calculus

Branch of Calculus

Rate of Change

Differentiation

- The rate of change
- $f(x)$, $\frac{dy}{dx}$, $\frac{d}{dx}$
- Single Time.

Successive Differentiation

- Multiple differentiations
- $f''(x)$, $\frac{d^2y}{dx^2}$, $\frac{d}{dx^2}$

Complex Numbers.

Techniques:- → constant

i) $\frac{d}{dx}[c] = 0$

ii) $\frac{d}{dx}[x] = 1$

iii) $\frac{d}{dx}[x^n] = n x^{n-1}$

iv) $\frac{d}{dx}[f(x) \pm g(x)] = \frac{d}{dx} f(x) \pm \frac{d}{dx} g(x)$

v) $\frac{d}{dx}[f(x) \cdot g(x)] = f(x) \cdot \frac{d}{dx} g(x) + \frac{d}{dx} f(x) \cdot g(x)$

vi) $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x) \frac{d}{dx}(f(x)) - f(x) \cdot \frac{d}{dx} g(x)}{(g(x))^2}$

vii) $f(g(x))$ → chain rule
$= f'g(x) \cdot g'(x)$

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Trigonometric Derivatives.

$\sin x' = \cos x$ $\quad\quad$ $\sin 0° = 0$

$\cos x' = -\sin x$ $\quad\quad$ $\cos 0° = 1$

$\tan x' = \sec^2 x$ $\quad\quad$ $\sec 0° = 1$

$\sec x' = \sec x \cdot \tan x$ $\quad\quad$ $\tan 0° = 0$

$\cot x' = -\operatorname{cosec}^2 x$ $\quad\quad$ $\cot 0° = \infty$

$\operatorname{cosec} x' = -\cot x \cdot \operatorname{cosec} x$ $\quad\quad$ $\operatorname{cosec} 0° = \infty$.

* L'Hôpitals

↳ Differentiation Technique

if $f(x) \to 0$, $g(x) \to 0$ $\quad \dfrac{f(x)}{g(x)} = \dfrac{0}{0}$

Here we apply derivative $\left.\dfrac{f'(x)}{g'(x)}\right]$ individual

if $f(x) \to \infty$, $g(x) \to \infty$ $\quad \dfrac{f'(x)}{g'(x)}$

$e^x = e^0 = 1$

$e^{x}{}' = e^x$

$e^{-x}{}' = -e^{-x}$

$\infty^{a>0} = \infty$ $\qquad e^{\infty} = \infty$

$\infty^{a<0} = 0$ $\qquad e^{-\infty} = 0$

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Q. $\lim_{x \to 0} \frac{e^{x}-1}{x^3}$ $\quad \frac{e^{0}-1}{0^3} = \text{indef}$

$\frac{e^x - 0}{3x^2} = \frac{e^x}{3x^2} = \frac{1}{0} = \infty$ Ans

Q. $\lim_{x \to \infty} \frac{x^{-4/3}}{\sin(1/x)} = \frac{\frac{1}{x^{4/3}}}{\sin 1/x} = \frac{1/\infty}{\sin 0} = \frac{0}{0}$ Indeterminant

$= \frac{-4/3\ x^{-4/3-1}}{\cos(1/x)(1/x^2)}$

$= \frac{-4/3\ x^{-7/3}}{\cos(1/x)(1/x^2)} = \frac{0}{0}$ Indeterminant

$\Rightarrow \frac{4/3\ x^{-7/3} \cdot x^2}{\cos(1/x)}$

$= \frac{4/3\ x^{-7/3+2}}{\cos(1/x)} = \frac{4/3\ x^{-1/3}}{\cos(1/x)} = \frac{\infty^{-1/3}}{1} = \frac{0}{1} = 0$ Ans.

Q. $\lim_{x \to 0} \frac{\ln x}{\operatorname{cosec} x}$ $\quad \frac{\infty}{\infty}$ = Indet.

$= \frac{1/x}{-\operatorname{cosec} x \cdot \cot x} = \frac{\cancel{x}\ 1}{-x \operatorname{cosec} x \cot x}$

$= \frac{\sin x \cdot \sin x}{-x \cos x}$

$= \frac{0 \times 0}{0 \times 1} = 0$ Ans

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Q3. $\lim_{x\to 0} \dfrac{e^{x}-e^{-x}}{\sin x}$ $\dfrac{0}{0}$ Indet.

$\dfrac{e^{x}+e^{x}}{\cos x} = \dfrac{2}{1} = 2$ Ans

Q4. $\lim_{x\to 0} \dfrac{e^{x^2}-1}{\cos x - 1}$ $\dfrac{0}{0}$ Indet.

$= \dfrac{2x e^{x^2}}{-\sin x} = \dfrac{0}{0}$ Indet.

$= \dfrac{2[x \cdot e^{x^2}]}{-\sin x}$

$= \dfrac{2[\cancel{x \cdot 2x e^{x^2}} + 1 \cdot e^{x^2}]}{-\cos x}$ $\to 0$

$= \dfrac{2(1)}{-1} = -2$ Ans

Q5. $\lim_{x\to 0} \dfrac{\tan x - \sin x}{x^2 \tan x}$

$\dfrac{\sin x/\cos x - \sin x}{x^2 \cdot \sin x/\cos x} = \dfrac{(\sin x - \sin x\cos x)/\cancel{\cos x}}{x^2 \cdot \sin x \cancel{\cos x}}$

$= \dfrac{\sin x - \sin x \cos x}{x^2 \cdot \sin x}$

$= \dfrac{\cancel{\sin x}(1-\cos x)}{x^2 \cdot \cancel{\sin x}}$

$\Rightarrow \dfrac{0+\sin x}{2x} = \dfrac{0}{0} \Leftarrow \dfrac{1-\cos x}{x^2}$ # Hugs

$\Rightarrow \dfrac{\cos x}{2} = \dfrac{1}{2} \longrightarrow$ Ans.

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Q6. $\lim_{x \to 0} \dfrac{x - \tan x}{x - \sin x}$

$\dfrac{x - \sin x / \cos x}{x - \sin x} = \dfrac{0}{0}$

$\Rightarrow \dfrac{1 - \sec^2 x}{1 - \cos x} = \dfrac{0}{0}$

$\Rightarrow \dfrac{0 - 2\sec x(\sec x \cdot \tan x)}{0 + \sin x} = \dfrac{0}{0}$

$\Rightarrow \dfrac{-2\sec^2 x \tan x}{\cos x} = \dfrac{}{1}$

$(\sec x \cdot \tan$

$\Rightarrow 2\left[\dfrac{\sec^2 x \cdot \sec^2 x + 2\sec x \cdot \tan x}{\cos x}\right.$

$\Rightarrow -2\left[\dfrac{\sec^4 x + 2\sec^2 x \cdot \tan^2 x}{\cos x}\right]$

$\Rightarrow -2$ Ans.

Q7. $\lim_{x \to 0} \dfrac{\ln(\sin 3x)}{\ln(\sin x)}$ $\dfrac{0}{0}$ Indet.

$\dfrac{1/\sin 3x}{1/\sin x} = \dfrac{\sin x}{\sin(3x)} = \dfrac{0}{0}$ Indet.

$= \dfrac{\cos(x)}{3\cos(3x)}$

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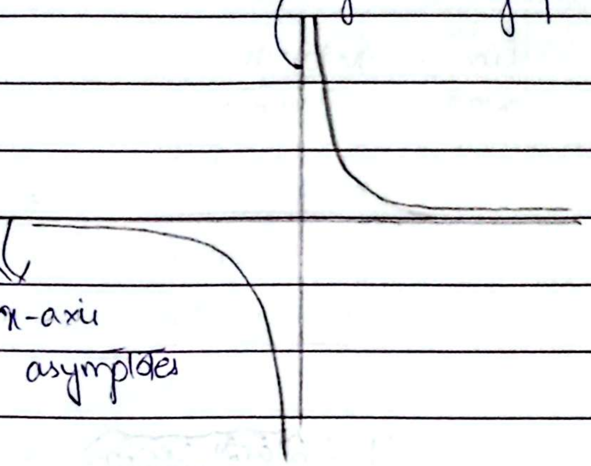
Asymptotes

Date: 30/01

$f(x) = \frac{1}{x} \quad \frac{1}{0} = \infty$



$x \neq 0$

$f(x) \neq 0$

Vertical Asymptote :- (Deno = 0)

$\lim_{x \to \pm a} f(x) = \mp \infty$

Horizontal Asymptote :-

$\lim_{x \to \pm a} f(x) = a$

Q. $\frac{2x+1}{x-3}$

$x - 3 = 0$

$x = 3$ vert. as.

$\lim_{x \to 3} \infty$

Horizontal :- numerator's power, denominator's power

$n < m$ (Horiz.)

$n = m$

$\lim_{x \to 0} \frac{2x+1}{x-3} \left(\frac{\infty}{\infty}\right) \quad \frac{x(2 + 1/x)}{x(1 - 3/x)} = 2$ Ans

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Q. $\dfrac{5x^0}{x^2+1}$

$0<2 \rightarrow d/dx \Rightarrow \dfrac{0}{2x}$

$= 0$

$\downarrow$

Horizontal asymptote

$x^2+1=0$

$x^2=-1$

$x=i \rightarrow$ No vertical asymptotes exist

Q. $\dfrac{x^2+1}{x}$

$2<1$

No horizontal asymptotes exist

$x=0$

$\hookrightarrow$ vertical asymptote

Inclined :-

- Convert into straight line eq
- Find the fraction
- Put $\lim_{x\to\infty}$

Q. $\dfrac{2x^2+3x+5}{x}$ $=0$ v. asymptote exist, horiz. asymptote ✗

$y = 2x + 3 + \dfrac{5}{x} \quad x \to \infty$

$y = 2x + 3 + 0$

$y = 2x + 3$

Leibnitz Theorem

Date: 04/02

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u v

$\frac{n!}{r(n-r)!}$

$\frac{dy^n}{dx^n}$ → no. of derivative

$(uv)^n$

$(fg)^n = f^n g^0 + {}^nC_1 f^{n-1} g^1 + {}^nC_2 f^{n-2} g^2 + {}^nC_3 f^{n-3} g^3 + \ldots$

Q. Find $\frac{d^4y}{dx^4}$ of $x^{4} \cdot \sin x$

		$'$	$''$	$'''$	$''''$
$f =$	x^4	$4x^3$	$12x^2$	$24x$	24
$g =$	$\sin x$	$\cos x$	$-\sin x$	$-\cos x$	$\sin x$

$(x^4 \cdot \sin x)^4 = \cancel{f} f'''' g^0 + {}^4C_1 f''' g' + {}^4C_2 f'' g'' + {}^4C_3 f' g''' + {}^4C_4 f^0 g''''$

$(24)(\sin x) + (4)(24x)(\cos x) + (6)(12x^2)(-\sin x) + (4)(4x^3)(-\cos x) + (1)(x^4)(\sin x)$

$= 24\sin x + 96x \cdot \cos x - 72x^2 \cdot \sin x - 16x^3 \cdot \cos x + x^4 \cdot \sin x$

Q. Find $\frac{d^2y}{dx^2}$ of $3x^2e^x + 2xe^x \Rightarrow \underset{f}{e^x}\underset{g}{(3x^2+2x)}$

f	e^x	e^x	e^x
g	$3x^2+2x$	$6x+2$	6

$(e^x(3x^2+2x))^2 = {}^2C_0 f'' g^0 + {}^2C_1 f' g' + {}^2C_2 f^0 g''$

$= (1)(e^x)(3x^2+2x) + (2)(e^x)(6x+2) + (1)(e^x)(6)$

$= e^x \cdot (3x^2+2x) + 2e^x \cdot (6x+2) + 6e^x$

$= e^x(3x^2+2x+2(6x+2)+6)$

$= e^x(3x^2+14x+10)$

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Q. find $\frac{d^6y}{dx^6}$ for $\overset{f}{e^{2x}}\,\overset{g}{\sin x}$

$f = e^{2x}$	$g = \sin x$	1
$f' = 2e^{2x}$	$g' = \cos x$	6
$f'' = 4e^{2x}$	$g'' = -\sin x$	15
$f''' = 8e^{2x}$	$g''' = -\cos x$	20
$f^{iv} = 16e^{2x}$	$g^{iv} = \sin x$	15
$f^{v} = 32e^{2x}$	$g^{v} = \cos x$	6
$f^{vi}\ 64e^{2x}$	$g^{vi} = -\sin x$	1

$(e^{2x}.\ \sin x)^6 = (64e^{2x})(\sin x) + {}^6C_1(32e^{2x})(\cos x) + {}^6C_2(16e^{2x})(-\sin x) +$
${}^6C_3(8e^{2x})(-\cos x) + {}^6C_4(4e^{2x})(\sin x) + {}^6C_5(2e^{2x})(\cos x) +$
${}^6C_6(e^{2x})(-\sin x)$

$= 64e^{2x}.\sin x + 192\ e^{2x}.\cos x - 240e^{2x}.\sin x$
$-160e^{2x}.\cos x + 60e^{2x}.\sin x + 12e^{2x}.\cos x - e^{2x}.\sin x$

$= e^{2x}(64\sin x + 192\cos x - 240\sin x - 160\cos x + 60\sin x + 12\cos x$
$- \sin x)$

$= e^{2x}(-117\sin x + 44\cos x)$

Taylor's & Maclaurin Series

CAG balls { Half class 11:40 enter

Date: 04/02

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special type of taylor a=0

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \cdots$$

$$f(x) = \underbrace{f(a)}_{\text{value of function}} + \underbrace{(x-a)f'(a)}_{\text{slope}} + \underbrace{\frac{(x-a)^2}{2!} f''(a)}_{\text{curvature}} + \underbrace{\frac{(x-a)^3}{3!} f'''(a)}_{\text{Higher figure definition}} + \cdots$$

$$\sum_{n=0}^{\infty} \frac{(x-a)^n}{n!} f^n(a)$$

Q. $\ln(1-x)$ by M.S. at least 4 derivatives

$$f(x) = \ln(1-x) = 1/(1-x)$$

$$f'(x) = \frac{-1}{1-x}$$

$$f''(x) = \frac{-1}{(1-x)^2}$$

$$f'''(x) = \frac{-2}{(1-x)^3}$$

$$f^{iv}(x) = \frac{-6}{(1-x)^4}$$

$f(x) =$

Taylor's Series and Maclaurin Series

Date: 06/02/26

$\downarrow$

power series if $a \neq 0$

$x + x^2 + x^3 + x^4 + \cdots$

Q. Find the power series of $f(x) = \frac{1}{x}$

at $x = 1$ —— $x - 1 = 0$

$\uparrow$

$\Rightarrow f(x) = \frac{1}{x} = 1$ $\quad (x-a)$; $a = 1$

$\Rightarrow f'(x) = -\frac{1}{x^2} = -1$ $\quad \Rightarrow f(x) = f(a) + (x-a) f'(a) + \frac{(x-a)^2}{2!} f''(a) + \cdots$

$\Rightarrow f''(x) = \frac{2}{x^3} = 2$ $\quad \Rightarrow f(x) = \frac{1}{x} + (x-1) \frac{-1}{x^2} + \frac{(x-1)^2}{2!} \cdot \frac{2}{x^3} +$

$\Rightarrow f'''(x) = \frac{-6}{x^4} = -6$ $\quad \frac{(x-1)^3}{3!} \cdot \frac{-6}{x^4} + \frac{(x-1)^4}{4!} \cdot \frac{24}{x^5}$

$\Rightarrow f^{iv}(x) = \frac{24}{x^5} = 24$ $\quad = 1 + (x-1)(-1) + \frac{(x-1)^2 (2)}{2!} + \frac{(x-1)^3 (6)}{3!} +$

$\frac{(x-1)^4 (24)}{4!}$

$= 1 - (x-1) + (x-1)^2 + (x-1)^3 + (x-1)^4 + \cdots$

Ans.

$K \rightarrow$ Kappa $\qquad \rightarrow R$

Curvature & Radius of Curvature of a Curve

Date: 06 - Feb - 26
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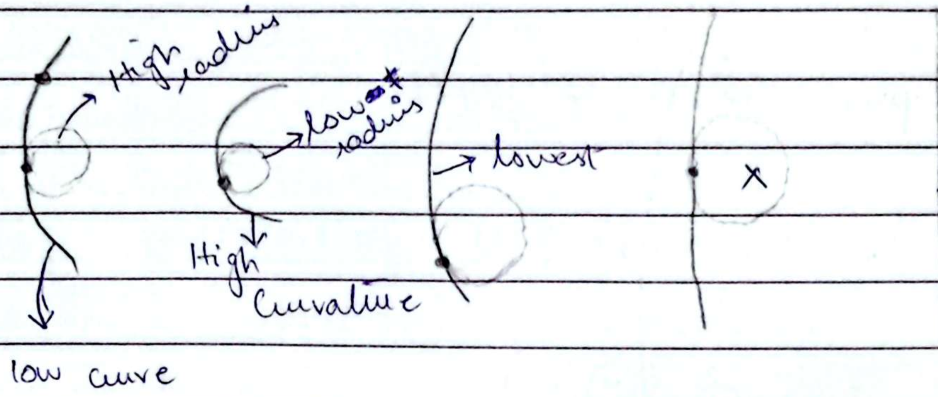

→ straight line
no curvature
→ Infinite / No radius

low curvature
↓
Measure curve
|
sharply bend

* How sharp the curve is bend on a particular point

* Inverse relation

$K = \frac{1}{R}$ ($K \rightarrow$ Curvature, $R \rightarrow$ Radius) $\qquad R = \frac{1}{K}$ ($K \rightarrow$ Curvature)

Cartesian ($y = x$)

$$R = \frac{\left[1 + (y')\right]^{3/2}}{|y''|}$$

Parametric

$x = x(t)$
$y = y(t)$

$$R = \frac{\left[(x')^2 + (y')^2\right]^{3/2}}{|x'y'' - y'x''|}$$

Polar (r, θ)

$r \rightarrow$ Magnitude, $\theta \rightarrow$ angle

$$R = \frac{\left[r^2 + (r')^2\right]^{3/2}}{|r^2 + 2(r')^2 - r.r''|}$$

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Q. Find radius of curvature (R) $y = c\cosh\left(\frac{x}{c}\right)$ (c → constant)

$$R = \frac{\left[1+\left(\sinh\left(x/c\right)\right)^2\right]^{3/2}}{\left|\cosh\left(\frac{x}{c}\right)\cdot\frac{1}{c}\right|}$$

$$y' = \cancel{c}\left[\sinh\left(\frac{x}{c}\right)\cdot\frac{1}{\cancel{c}}\right]$$

$$y'' = \cosh\left(\frac{x}{c}\right)\cdot\frac{1}{c}$$

$$R = \frac{\left(\cosh^2(x/c)\right)^{3/2}}{\frac{1}{c}\cdot\cosh\left(\frac{x}{c}\right)}$$

$$\left[\frac{y}{c}\right]^2 = \left(c\cdot\cosh^2\left(\frac{x}{c}\right)\right)$$

$$R = \frac{\cosh^{\cancel{3}\,2}(x/c)}{\frac{1}{c}\cdot\cancel{\cosh\left(\frac{x}{c}\right)}}$$

$$R = c\cdot\cosh^2\left(\frac{x}{c}\right)$$

$$R = c\cdot\frac{y^2}{c^2}$$

$$R = \frac{y^2}{c}$$ Ans

Q2. $x = a(\cos t + t\sin t)$
$y = a(\sin t - t\cos t)$ $\quad a > 0$

$\frac{dx'}{dt}$

Numerator :- $\left[(a(t\cos t))^2 + (a(t\sin t))^2\right]^{3/2}$

$\Rightarrow \left(a^2t^2\cos^2 t + a^2t^2\sin^2 t\right)^{3/2}$

$\Rightarrow \left(a^2t^2(\cos^2 t + \sin^2 t)\right)^{3/2}$

$\Rightarrow (a^2t^2)^{3/2} = a^3t^3$

$x' = a[\cancel{-\sin t} + (t\cos t + \cancel{\sin t})]$

$x' = a(t\cos t)$

$y' = a[\cancel{\cos t} - (t\sin t + \cancel{\cos t})]$

$y' = a(t\sin t)$

$x'' = a[-t\sin t + \cos t]$

$y'' = a[\sin t + t\cos t]$

Denominator :- $|(a(t\cos t))(a[\sin t + t\cos t]) - ((a t\sin t)(a(-t\sin t + \cos t))|$

$\Rightarrow |a^2\cancel{t\cos t\sin t} + a^2t^2\cos^2 t + a^2t^2\sin^2 t - a^2\cancel{t\sin t\cos t}|$

$= a^2t^2\cos^2 t + a^2t^2\sin^2 t$

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$|a^2t^2(\cos^2t+\sin^2t)|$

$= a^2t^2$

finally, $\dfrac{a^3t^3}{a^2t^2} = at$ Ans.

Q3. $r\theta = a$

$r = \dfrac{a}{\theta}$

$r' = a\theta^{-1}$

$= \dfrac{-1\times a}{\theta^2} = \dfrac{-a}{\theta^2}$

$r'' = -a\theta^{-2}$

$= \dfrac{-2\times -a}{\theta^3}$

$= \dfrac{2a}{\theta^3}$

$= \dfrac{\left[(a/\theta)^2+(-a/\theta^2)^2\right]^{3/2}}{|(a/\theta)^2+2(-a/\theta^2)^2-(a/\theta)(2a/\theta^3)|}$

Numerator $= \left(\dfrac{a^2}{\theta^2}+\dfrac{a^2}{\theta^4}\right)^{3/2}$

$= \left(\dfrac{a^2\theta^2+a^2}{\theta^4}\right)^{3/2}$

$= \left(\dfrac{a^2(\theta^2+1)}{\theta^4}\right)^{3/2} = \dfrac{a^3(1+\theta^2)^{3/2}}{\theta^6}$

Denominator $= \left|\dfrac{a^2}{\theta^2}+2\left(\dfrac{a^2}{\theta^4}\right)-\dfrac{a}{\theta}\cdot\dfrac{2a}{\theta^3}\right|$

$= \left|\dfrac{a^2}{\theta^2}+\cancel{\dfrac{2a^2}{\theta^4}}-\cancel{\dfrac{2a^2}{\theta^4}}\right|$

$= \left|\dfrac{a^2}{\theta^2}\right|$

$\Rightarrow \dfrac{\dfrac{\cancel{a^3}^{\,1}(1+\theta^2)^{3/2}}{\cancel{\theta^6}^{\,4}}}{\cancel{a^2}/\cancel{\theta^2}} = \dfrac{a(1+\theta^2)^{3/2}}{\theta^4}$ Ans.

Q4. $r = 2\cos 2\theta$ at $\theta = \dfrac{\pi}{12}$

Numerator : $\cancel{\left((2\cos 2\theta)^2 + (-4\sin 2\theta)^2\right)^{3/2}}$

$\Rightarrow \cancel{\left(4\cos^2 2\theta + 16\sin^2 2\theta\right)^{3/2}}$

$\Rightarrow$

$r = 2\cos 2\theta$

[illegible]

$r' = -4\sin 2\theta = -2$

$r'' = -8\cos 2\theta = -4\sqrt{3}$

Date: 13-Feb-26

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Q. If $r = a(1+\cos\theta)$ then prove that: $\frac{\rho^2}{r} = \frac{8a}{9}$ → Question

→ radius of curvature

$$\frac{[r^2 + (r')^2]^{3/2}}{|r^2 + 2(r')^2 - rr''|}$$

$r' = a(-\sin\theta)$

$r'' = -a\cos\theta$

$$\Rightarrow \frac{[(1+\cos\theta)^2 + (a(-\sin\theta))^2]^{3/2}}{|\cancel{a}(1+\cos\theta))^2 + 2(a(-\sin\theta))^2 - (1+\cos\theta)(-a\cos\theta)}$$

$$\Rightarrow \cancel{\frac{(1+2\cos\theta+\cos^2\theta+a^2\sin^2\theta)^{3/2}}{|1+2\cos\theta+\cos^2\theta+2}}$$

$$\Rightarrow \frac{[(a(1+\cos\theta))^2 + (a(-\sin\theta))^2]^{3/2}}{|(a+a\cos\theta)^2 + 2(a(-\sin\theta))^2 - (a+a\cos\theta)(-a\cos\theta)|}$$

$$\Rightarrow \frac{(a^2 + 2a^2\cos\theta + a^2\cos^2\theta + a^2\sin^2\theta)^{3/2}}{|a^2 + 2a^2\cos\theta + 2a^2\cos^2\theta + 2a^2\sin^2\theta + a^2\cos\theta + a^2\cos^2\theta|}$$

$$\Rightarrow \frac{(a^2 + 2a^2\cos\theta + a^2)^{3/2}}{a^2 + 2a^2\cos\theta + 2a^2\cos^2\theta + 2a^2\sin^2\theta + a^2\cos\theta}$$

N: $(2a^2(1+\cos\theta))^{3/2} = 8a^3(1+\cos\theta)^{3/2}$

D:- $a^2 + 2a^2\cos\theta + 2a^2 + a^2\cos\theta$

$= 3a^2 + 3a^2\cos\theta$

$= 3a^2(1+\cos\theta)$

$$= \frac{8a^{\cancel{3}}(1+\cos\theta)^{3/2}}{3\cancel{a^2}(1+\cos\theta)} = \frac{8a(1+\cos\theta)^{1/2}}{3} = \frac{2\sqrt{2a}\sqrt{1+\cos\theta}}{3}$$

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$$\frac{\rho^2}{r} = \frac{\left(\dfrac{2\sqrt{2}\,a\sqrt{1+\cos\theta}}{3}\right)^2}{a(1+\cos\theta)}$$

$$= \frac{\left(4\times 2\times a^{\cancel{2}}\cancel{(1+\cos\theta)}\right)/9}{\cancel{a(1+\cos\theta)}}$$

$$= \frac{8a}{9} \quad \underline{\text{Ans}}.$$

∂ — sigma

PARTIAL DIFFERENTIATION — six parts

Date: 13/02

M T W T F S S

Differentiation

Ordinary — Derivation with respect to one independent variable

Partial — Derivation w.r.t more than one independent variables

$\frac{dy}{dx}, \frac{d^2y}{dx^2} \ldots$

$z = (x, y)$

Dep — Indep.

→ Six Parts:

$z = x^2 + y^2$

$\frac{\partial z}{\partial x}, f_x, U_x, P$

$z = 2x + 0 = 2x$

$\frac{\partial z}{\partial y} = f_y = U_y = q$

$\frac{\partial z}{\partial y} = 2y$

$\frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial x}\right) = U_{xx} = f_{xx} = r$

$= 2$

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$$\frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial y}\right) = t$$

$$= z$$

$$\left.\begin{aligned} f_{xy} &= \frac{\partial}{\partial y}\left(\frac{\partial z}{\partial x}\right) = \frac{\partial^2 z}{\partial y \partial x} \\ f_{yx} &= \frac{\partial}{\partial x}\left(\frac{\partial z}{\partial y}\right) = \frac{\partial^2 z}{\partial x \partial y} \end{aligned}\right\} = s$$

Q1. $z = 2x^2 + 3xy + 6x + 7y$

Q2. $z = e^x \sin y + xy^2$

Q3. $z = x^2y^3 + 3x^2\sin^2 y + e^y \cos^2 x$

1. $\frac{\partial z}{\partial x} = 4x + 3y + 6$

$$\left.\begin{aligned} f_{xy} &= 3 \\ f_{yx} &= 3 \end{aligned}\right\} \text{Ans}$$

$\frac{\partial z}{\partial y} = 3x + 7$

$\frac{\partial^2 z}{\partial x^2} = 4$

$\frac{\partial z}{\partial y^2} = 0$

P

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2. $\frac{\partial z}{\partial x} = e^x \sin y + y^2$

$\frac{\partial z}{\partial y} = e^x \cos y + 2xy$

$\frac{\partial^2 z}{\partial x^2} = e^x \sin y$

$\frac{\partial^2 z}{\partial y^2} = -e^x \sin y + 2x$

$f_{xy} = e^x \cos y + 2y$

$f_{yx} = e^x \cos y + 2y$

J

3. $\frac{\partial z}{\partial x} = 2xy^3 + 6x \sin y - e^y \sin 2x$

$\frac{\partial z}{\partial y} = 3x^2y^2 + 3x^2 \cos y + e^y \cos^2 x$

$\frac{\partial^2 z}{\partial x^2} = 2y^3 + 6\sin y - 2e^y \cos 2x$

$\frac{\partial^2 z}{\partial y^2} = 6x^2y - 3x^2 \sin y + e^y \cos^2 x$

$f_{xy} = 6xy^2 + 6x \cos y - e^y$

$f_{yx} =$

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$3z^2 - 3y \quad -3$

Q4. If $z^3 - 3yz - 3x = 0$

prove $zp = q$

→ Part D in term of x

$$3z^2 \cdot \frac{\partial z}{\partial x} - 3y\frac{\partial z}{\partial x} - 3 = 0$$

$$3z^2 \frac{\partial z}{\partial x} - 3y \frac{\partial z}{\partial x} = 3$$

$$\cancel{3}\frac{\partial z}{\partial x}\left(z^2 - y\right) = \cancel{3}$$

$$\frac{\partial z}{\partial x} = \frac{1}{z^2-y}$$

w.r.t 'y'

$$3z^2 \frac{\partial z}{\partial y} - 3\left[y\frac{\partial z}{\partial y} + z\right] = 0$$

$$3z^2 \frac{\partial z}{\partial y} - 3y\frac{\partial z}{\partial y} - 3z = 0$$

$$\cancel{3}\frac{\partial z}{\partial y}\left(z^2 - y\right) = \cancel{3}z$$

$$\frac{\partial z}{\partial y} = \frac{z}{z^2-y}$$

Partial Differentiation

Date: 18/02

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Reading

$$u = e^{xyz}$$

dep.

* z ka differential y mein and xy ka differential x mein

$$U_{xyx} = \frac{\partial^3 u}{\partial x \partial y \partial z}$$

$$\frac{du}{dz} = xy\, e^{xyz}$$

$$\frac{d}{dy}\left(\frac{du}{dz}\right) = xy \cdot xz e^{xyz} + x e^{xyz}$$

$$= x^2 yz e^{xyz} + x e^{xyz}$$

$$= e^{xyz}\left(x^2 yz + x\right)$$

$$\frac{d}{dx}\left(\frac{d^2u}{dy\,dz}\right) = e^{xyz}\left(2xyz + 1\right) + yz e^{xyz}\left(x^2yz + x\right)$$

$$= 2xyz e^{xyz} + e^{xyz} + x^2y^2z^2 e^{xyz} + xyz e^{xyz}$$

$$= e^{xyz}\left(3xyz e^{xyz} + e^{xyz} + x^2y^2z^2 e^{xyz}\right) \quad \text{Ans}$$

Extreme Value of Two Variable :-

Date: 18/02/26

Max — Min

② with constraints | helping | subject
(Two Functions)

① without constraints | subject | help
(Single Function)

1. f_x and f_y – Partial Differentiation
2. $f_x = 0$ and $f_y = 0$ – Equate to zero
   for stationary points $(x_1, y_1)(x_2, y_2)$ ...

$f_{xx} = A$ $\quad$ $D = (f_{xx})(f_{yy}) - (f_{xy})^2$

$f_{yy} = C$ $\quad$ $D = AC - B^2$

$f_{xy} = B$

Stationary Point	$A = f_{xx}$	$B = f_{xy}$	$C = f_{yy}$	$D = AC - B^2$
x_1, y_1	$A > 0$			$D > 0$
x_2, y_2	$A < 0$			$D < 0$
x_3, y_3	$A = 0$			$D = 0$
⋮				

if $A > 0$ and $D > 0$ Minimum

if $A < 0$ and $D > 0$ Maximum

if $D < 0$ (Saddle Point)

if $D = 0$ (Test fail | Inconclusive Test)

Date: M T W T F S S

Q1. Find the extreme values of the functions

i) ~~x~~ $f(x,y) = x^3 + y^3 - 3x - 12y + 20$

1) $f_x = 3x^2 - 3$
$f_y = 3y^2 - 12$

2) $f_x = 0$, $f_y = 0$
$3x^2 - 3 = 0$, $3y^2 - 12 = 0$
$3x^2 = 3$, $3y^2 = 12$
$x^2 = 1$, $y^2 = 4$
$x = \pm 1$, $y = \pm 2$

(1, 2)
(−1, 2)
(1, −2)
(−1, −2)

3) $f_{xx} = 6x$, $f_{xy} = 0$
$f_{yy} = 6y$, $f_{yx} = 0$

Stationary Point	~~$A = 6x$~~ $A = f_{xx}$	$B = 0$ $B = f_{xy}$	$C = 6y$ $C = f_{yy}$	$D = AC - B^2$	
(1, 2)	$A = 6 > 0$	$= 0$	$C = 12 > 0$	$D = 72 > 0$	Max
(−1, −2)	$A = -6 < 0$	$= 0$	$C = -12 < 0$	$D = 72 > 0$	
(1, −2)	$A = 6 > 0$	$= 0$	$C = -12 < 0$	$D = -72 < 0$	Saddle Point
(−1, 2)	$A = -6 < 0$	$= 0$	$C = 12 > 0$	$D = -72 < 0$	Saddle point

$f(1,2) = 2$ Min
$f(-1,-2) = 38$ Max.

Good

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Q2. $f(x,y) = x^4 + y^4 - 2x^2 + 4xy - 2y^2$

1) $f_x = 4x^3 - 4x + 4y$

$f_y = 4y^3 + 4x - 4y$

2) $f_x = 0$, $f_y = 0$

$4x^3 - \cancel{4x} + \cancel{4y} = 0$, $4y^3 + \cancel{4x} - \cancel{4y} = 0$

$4x^3 + 4y^3 = 0$

$4x^3 = -4y^3$

$x = -y$

$f_x \Rightarrow 4(-y)^3 - 4(-y) + 4y = 0$

$\Rightarrow -4y^3 + 4y + 4y = 0$

$\Rightarrow 8y - 4y^3 = 0$

$\Rightarrow -4y(2 - y^2) = 0$

$\Rightarrow -4y = 0$, $2 - y^2 = 0$

$y = 0$, $y^2 = 2$

$y = 0$, $y = \pm\sqrt{2}$

$x = 0$, $x = \mp\sqrt{2}$

$(0,0)$ $(+\sqrt{2}, -\sqrt{2})$ $(-\sqrt{2}, +\sqrt{2})$

3) $f_{xx} = 12x^2 - 4$

$f_{yy} = 12y^2 - 4$

$f_{xy} = +4$

$f_{yx} = +4$

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Stationary Point	$A = 12x^2 - 4$	$B = 4$	$C = 12y^2 - 4$	$D = AC - B^2$
$0, 0$	$A = -4 < 0$	4	$C = -4 < 0$	$D = 0$ (Test Fail)
$\sqrt{2}, -\sqrt{2}$	$A = 20 > 0$	4	$C = 20 > 0$	$D = 384$ Min
$-\sqrt{2}, \sqrt{2}$	$A = 20 > 0$	4	$C = 20 > 0$	$D = 384$ Min

$$f(-\sqrt{2}, \sqrt{2}) = -8$$
$$f(\sqrt{2}, -\sqrt{2}) = -8$$
Same

Ans

Q. $f(x,y) = x^2 - xy + y^2 - 2x + y$

1) $f_x = 2x - y - 2$

$f_y = -x + 2y + 1$

2) $f_x = 0$, $f_y = 0$

$2x - y - 2 = 0$, $-x + 2y + 1 = 0$

$$4x - 2y - 4 = 0$$
$$-x + 2y + 1 = 0$$
$$3x - 3 = 0$$

$x = 1$

$y = 0$

$(1, 0)$

3) $f_{xx} = 2$

$f_{yy} = 2$

$f_{yx} = -1$

$f_{xy} = -1$

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S.P	$A = 2$	$B = -1$	$C = 2$	$D = AC - B^2$
$(1, 0)$	$A = 2 > 0$	$B = -1 < 0$	$C = 2 > 0$	$D = 5$ - Min

$f(1,0) = -1$ Ans.

Q. $f(x,y) = x^3y^2(1-x-y)$

$x^3y^3 - x^4y^3 - x^3y^4$

1) $f_x = 3x^2y^3 - 4x^3y^3 - 3x^2y^4$

$f_y = 2x^3y^2 - 2x^4y^2 - 4x^3y^3$

2) $f_x = 0$, $f_y = 0$

$3x^2y^3 - 4x^3y^3 - 3x^2y^4 = 0$, $3x^3y^2 - 3x^4y^2 - 4x^3y^3 = 0$

$-3x^2y^3 + \cancel{4x^3y^3} + 3x^2y^4 = 0$, $3x^3y^2 - 3x^4y^2 - \cancel{4x^3y^3} = 0$

$-3x^2y^3 + 3x^2y^4 + 3x^3y^2 - 3x^4y^2 = 0$

$-3x^2y^2$

Date:
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Q. $f(x,y) = x^3y^2(1-x-y)$

$x^3y^2 - x^4y^2 - x^3y^3$

$f_x = 3x^2y^2 - 4x^3y^2 - 3x^2y^3 = 0$

$f_y = 2x^3y - 2x^4y - 3x^3y^2 = 0$

$f_x = x^2y^2(3-4x-3y) = 0$

$f_y = x^3y(2-2x-3y) = 0$

$x=0,\ y=0$

$f_{xx} = 6xy^2 - 12x^2y^2 - 6xy^3$

$f_{yy} = 6x^2y - 8x^3y - 9x^2y^2$

$f_{xy} = 6x^2y - 8x^3y - 9x^2y^2$

$f_{yx} = 6x^2y - 8x^3y - 9x^2y^2$

$3 - 4x - 3y = 0$

$-(2 - 2x - 3y = 0)$

$1 - 2x = 0$

$2x = 1$

$x = 1/2$

$y = 1/3$

S.P	$A = f_{xx}$	$B = f_{yx}$	$C = f_{yy}$	$D = AC - B^2$
0, 0	0	0	0	0 (test fail)
1/2, 1/3	$-1/9 < 0$	$-1/12 < 0$	$-1/12 < 0$	$D = 1/432$ Ans

$\boxed{1/432}$ Ans

Date:
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Q. Discuss the min and max of the function

$$f(x,y) = x^3 + y^3 - 3ayx$$

$f_x = 3x^2 - 3ay$

$f_y = 3y^2 - 3ax$

$3x^2 - 3ay = 0$

$3y^2 - 3ax = 0$

$3x^2 = 3ay$

$y = \frac{x^2}{a}$

$3y^2 - 3ax = 0$

$\left(\frac{x^2}{a}\right)^2 - ax = 0$

$\frac{x^4}{a^2} - ax = 0$

$x^4 - a^3x = 0$

$\cancel{x^4 = a^3x}$

$x(x^3 - a^3) = 0$

$x = 0, \; x = a$

$y = 0, \; ay = a$

$(0,0) \; (a,a)$

$f_{xx} = 6x$

$f_{yy} = 6y$

$f_{yx} = -3a$

$f_{xy} = -3a$

S.P	A f_{xx}	B f_{yx}	C f_{yy}	$D = AC - B^2$
0,0	0	$-3a$	0	$D = -9a^2$ Saddle point
a,a	$6a$	$-3a$	$6a$	$D = 36a^2 - 9a^2$ $= 27a^2$ Max, Min

Chapter 4: Sequence & Series (CLO-3)

Date: 18/02 M T W T F S S

1, 2, 3, 4 ---

↪ term's

$a_1, a_2, a_3, a_4, a_5 --- a_n$

$$\sum_{n=1}^{\infty} n = 1+2+3+4+5+6+7+8 -----$$

Divergence (Infinity)

Convergence (well-defined)

1) Comparision Test:

(P-series)

↳ Helping Test

DCT

Direct Comparision Test

LCT

Limit Comparision Test

+ve always $\begin{cases} U_n = \text{Question ki series} \\ V_n = \text{We take out from question} \end{cases}$

U_n given

V_n from U_n

$\lim_{n\to\infty} \frac{U_n}{V_n} = L$ → if defined → P series further
→ if infinity → Test fail

further.

$V_n = \frac{1}{n^p}$

$p > 1$ convergent

$p \leq 1$ divergent

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(2) $\sum_{K=1}^{\infty} \frac{3K^3 - 2K^2 + 4}{K^7 - K^3 + 2}$

$U_n = \frac{3K^3 - 2K^2 + 4}{K^7 - K^3 + 2}$

* Shortcut :- Fractionize the highest and lowest powers of numerator and denominator

$V_n = \frac{K^3}{K^7} = \frac{1}{K^4}$ or $\frac{3}{K^4}$

$\lim_{n \to \infty} \frac{U_n}{V_n} = \frac{3K^3 - 2K^2 + 4}{K^7 - K^3 + 2} \times \frac{K^4}{1}$

$= \frac{3K^7 - 2K^6 + 4K^4}{K^7 - K^3 + 2}$

$= \cancel{\frac{K^7}{K^7}} \quad \frac{K^7}{K^7}\left(\frac{3}{1}\right) = 3$ - defined

P - Series :-

$V_n = \frac{1}{K^4}$ → converge since $4 > 1$

Date: M T W T F S S

Q2. $\frac{5}{4} + \frac{7}{9} + \frac{9}{16} ---$

$$\sum_{n=1}^{\infty} \frac{2n+3}{(n+1)^2}$$

$$U_n = \frac{2n+3}{n^2+2n+1}$$

$$V_n = \frac{n}{n^2} = \frac{1}{n}$$

$$\lim_{n\to\infty} \frac{U_n}{V_n} = \frac{2n+3}{n^2+2n+1} \times n$$

$$= \frac{2n^2+3n}{n^2+2n+1} \quad \frac{n^2}{n^2}\left(\frac{2}{1}\right) = 2 \text{ - defined}$$

P- Series

$$\frac{1}{n} \longrightarrow$$